

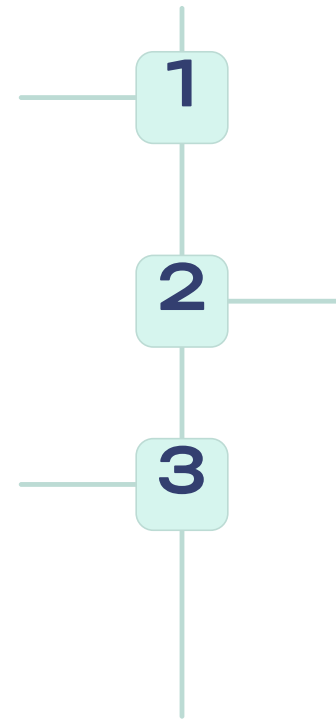


AI-Driven Agriculture Damage Prediction System

This presentation introduces an AI-driven agriculture damage prediction system, designed to revolutionize modern farming. By integrating AI, IoT sensors, satellite imagery, and blockchain, it empowers farmers with accurate forecasts and timely alerts to mitigate crop losses from pests, diseases, and extreme weather.

**By: 1.Harish
2.Rishank
3.Nithish**

Under Guidance Of : Richa Sharma



Purpose and Technical Workings

Purpose

The system aims to reduce crop losses, promote food production, and encourage sustainable practices by providing farmers with timely estimates and proactive responses to threats like pests, diseases, and extreme weather.

Technical Workings

- Multi-source data integration
- AI-powered prediction models
- Autonomous warnings and response systems
- Secure and transparent data storage
- Simple and available user interface



Unique Attributes



Future Intelligence



Automated Support



Scalability

The system constantly estimates future dangers, providing farmers with the ability to work prematurely and prevent damage. It processes raw data into actionable insights, protecting farmers without manual analysis. The modular structure allows implementation on smallholder farms or large corporate farms.



Problems Addressed

Unexpected Crop Failures

Agricultural losses due to pests, diseases, and climatic extremes often catch farmers by surprise, impacting yields and income.

Limited Manual Monitoring

Manual crop inspection is labor-intensive, time-consuming, and prone to human error, especially on large farms.

High Input Costs

Resources like water, pesticides, and fertilizers are frequently overused due to the lack of accurate data.

Objectives of the Invention



Increase Crop Protection Efficiency

Deliver accurate and timely predictions to enable proactive measures, reducing the impact of potential threats.



Minimize Financial Risk

Reduce unexpected agricultural damage by offering timely alerts and tailored recommendations.



Encourage Eco-Friendly Farming

Promote precision agriculture by recommending actions based on actual field data, reducing excessive use of resources.





State of the Art and Novelty

Study	Research Gap	Novelty
Crop Disease Prediction based on AI and ML Models (US20220414795A1)	May not fully integrate real-time environmental data for proactive disease prediction.	Combines real-time sensor data, climate information, and AI-driven image analysis for preemptive interventions.
Integrated IoT System for Smart Agriculture Management (US10728336B2)	Lacks personalized predictive recommendations tailored to individual farm conditions.	Feeds collected IoT data into AI models that generate forecasts and provide customized advice for each farmer's unique situation.



System Components

1

Sensor and Satellite-Based Data Infrastructure

IoT sensors and satellite/drone imaging provide real-time data on soil conditions and crop health.

2

Communication and Integration Tools

Wireless sensor communication and API integrations ensure seamless data updates and enriched analysis.

3

Central Intelligence Platform

AI-powered analytics engine, farmer interface, and blockchain-backed data logging for accurate insights and transparency.

Results and Advantages

1

Boost in Farming Efficiency

Early risk detection and smarter farm decisions.

2

Better Use of Resources

Targeted input usage and eco-friendly practices.

3

Enhanced Support for Farmers

Live alerts, advice, and accessible design.

The AI-driven system offers a smarter, more reliable, and efficient way to manage agricultural risks. It provides early, accurate, and actionable insights, supporting smarter farming, lowering input waste, and encouraging long-term sustainability.

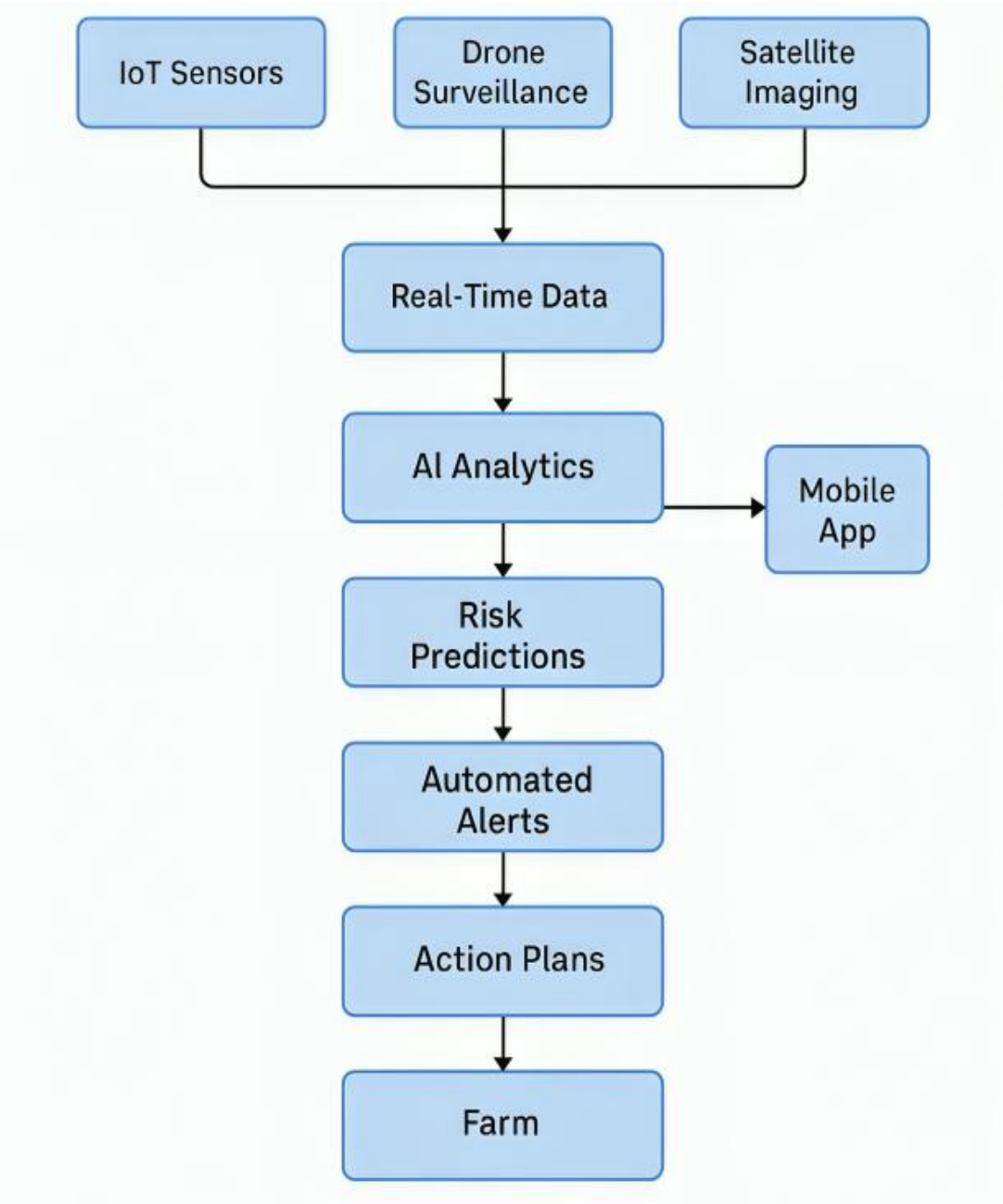
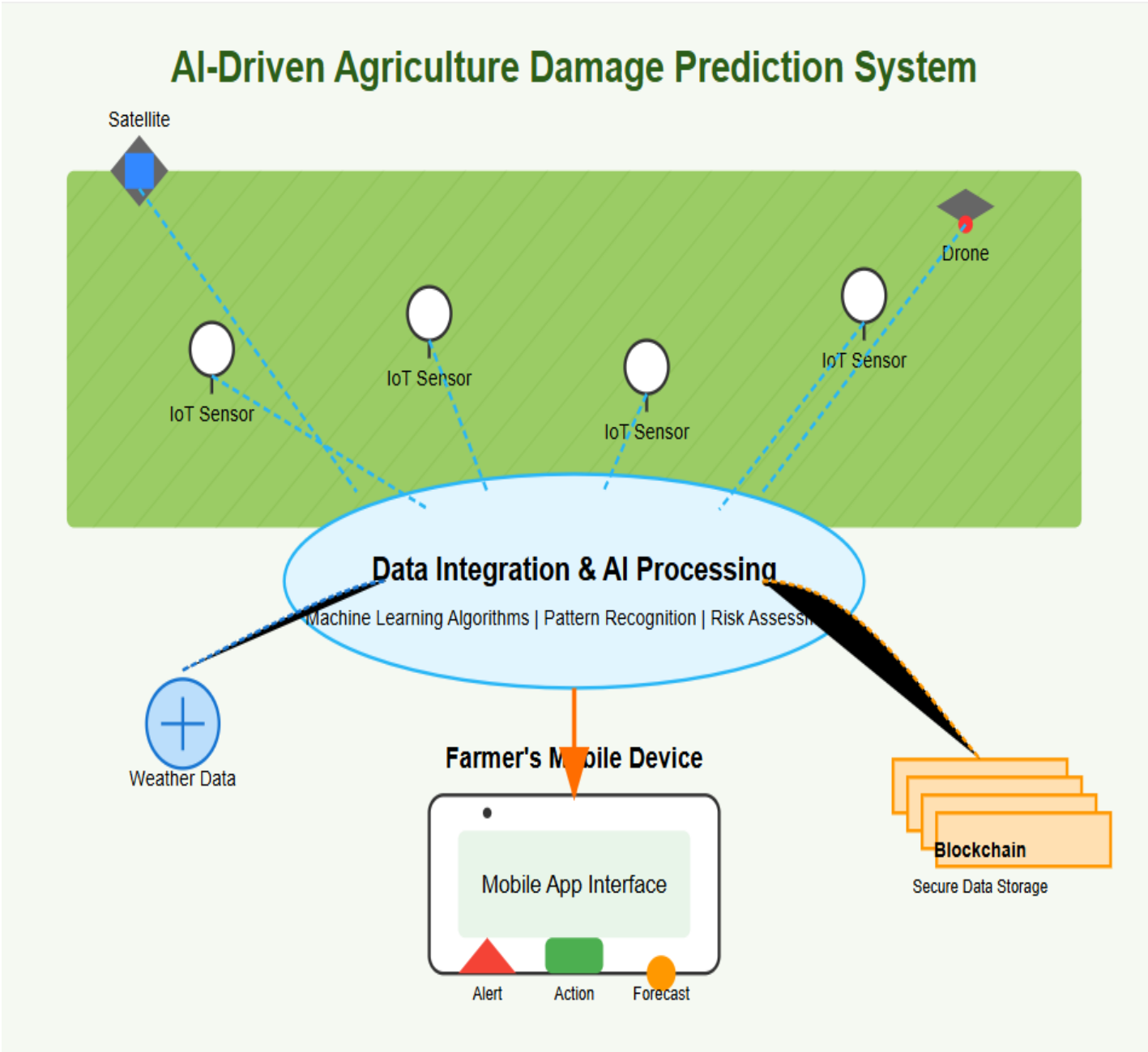


Potential for Commercialization

The system has the potential for commercialization through partnerships with companies specializing in AI-powered agriculture and IoT for smart farming. Potential companies to contact for commercialization include CropX Technologies and AgEagle Aerial Systems Inc.

The project is currently in the idea and planning stage, with a working prototype expected in about a year.

Data Flow Diagram (DFD)



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EXPANSION:

To bring this idea to life and ensure it's useful across diverse agricultural settings, several key aspects must be considered during development and rollout:

1. Compatibility with Crops and Locations

Tailored AI Models: Each crop has unique risks. The system must be trained on region-specific data to give accurate predictions.

Adaptable to Local Conditions: Soil, weather, and farming styles vary—models should reflect those local realities.

2. Hardware and Sensor Setup

Good Coverage: Sensors must be placed strategically to reflect conditions across the whole field.

Built for the Outdoors: Devices should be tough enough to work in rain, heat, or dust without

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DETAILED DESCRIPTION:

The AI-driven prediction system for agricultural injuries is an ideological innovative solution aimed at bringing revolution in modern agriculture by predicting possible crop injuries through the use of AI and smart technologies. It is designed to strengthen farmers with accurate forecasts and timely alerts that allow them to prevent or reduce damage caused by pests, diseases, soil falls and extreme weather conditions. System AI-based data integrates analyzes, IoT sensor, satellite images and blockchain technology into the integrated future platform.

2. System Components

2.1 Sensor and Satellite-Based Data Infrastructure

IoT Sensors: He was deployed on the farm to monitor important parameters such as soil moisture, temperature, humidity and nutritional level. These sensors continuously send real -time data to analysis platforms.

Satellite and Drone Imaging: Provides visual data on plant health, crop coverage and early signs of disease or water stress. This enables the assessment of the area on a large scale and improves data granularity.

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